

Formula 1

A Mathematical Analysis to Identify the Top Formula 1 Driver in History

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1. Introduction

When posed the question, “Who is or was the best Formula 1 driver of all time?” an AI agent will likely first mention **Lewis Hamilton**, followed by **Michael Schumacher**, based primarily on their respective world championship titles. If asked for a list of top drivers, it might also include **Juan Manuel Fangio**, **Alain Prost**, **Max Verstappen**, and **Sebastian Vettel**, arranged alphabetically. **Fangio** stands out with five championships, while the others each have four.

Interestingly, many fans regard **Ayrton Senna**, with his three titles, as a superior driver compared to **Alain Prost**, despite the latter’s four championships. This raises the question: how significant are titles in evaluating a driver’s greatness? Should we consider alternative criteria for determining who truly excels?

If we delve deeper, **Juan Manuel Fangio** won five titles in the 1950s, during the formative years of the Formula 1 World Championship. His legacy is so profound that his name has become synonymous with exceptional driving. In French-speaking regions, the phrase “driving like Fangio” denotes flawless driving. A notable reference appears in *The Adventures of Tintin: The Calculus Affair* [1], where a character remarks, “This must be Fangio at the wheel” as a car skillfully avoids a net, reflecting Fangio’s prowess during the height of his career. (In the English edition, it is translated as “They must have a Jack Brabham at the wheel!”)

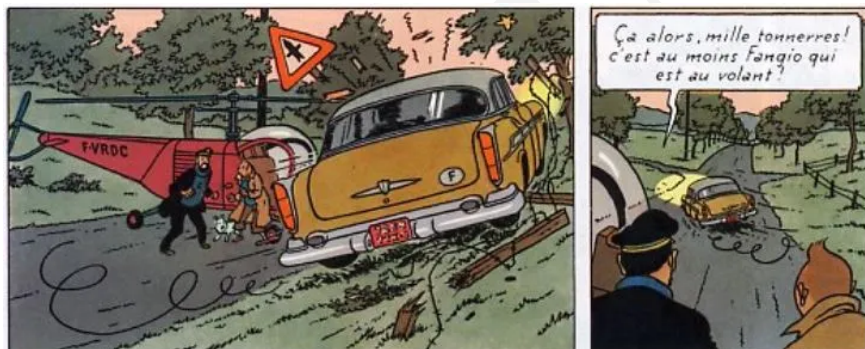


Figure 1: “This must be Fangio at the wheel!”

Complicating matters further, the rules of Formula 1 have evolved significantly over the years. The number of races has increased (from seven during Fangio’s era to 22 today), and the points system has undergone various changes. For instance, the winner received 8 points until **1960**, then 9 points until **1990**, followed by 10 points until **2009**, and has been set at 25 points since **2010**. Additionally, until **1990**, only the best results counted towards the championship, which allowed Senna to secure the title in **1990** against Prost, despite the latter having more points overall; under modern rules, Prost would have been awarded the title instead.

Ultimately, it is evident that even the most skilled drivers depend significantly on their cars’ performance. Both Alain Prost in **1991** and Lewis Hamilton in **2025** encountered substantial difficulties while driving for Ferrari, so much so that Prost was dismissed from his team after comparing his car to a “truck.” This raises an important question: to what extent is a world title attributable to the car, and how much is a reflection of the driver’s ability?

2. Dataset

2.1 Source

The Formula 1 data, including information on drivers, teams, qualifications, and results, is readily available online, such as on Wikipedia [2]. However, for research purposes, a well-maintained dataset may be more valuable. A suitable dataset can be found on Kaggle [3], although accessing it requires an account that some users may not possess. A publicly accessible version of this dataset is also available on GitHub Training Insights, but to guarantee its ongoing accessibility, I have uploaded a copy to my GitHub repository [4]. This redundancy will ensure that the dataset remains available even if the original site removes it.

2.2 Files

The dataset is organized as nineteen different files : 18 comma-separated value files and a json file. We will only use the 9 files below :

Dataset Files	
	circuits.csv
	constructor_results.csv
	constructor_standings.csv
	constructors.csv
	driver_standings.csv
	drivers.csv
	races.csv
	results.csv
	status.csv

The other files listed below aren't relevant for this study :

Dataset Files	
	fatal_accidents_drivers.csv
	fatal_accidents_marshalls.csv
	lap_times.csv
	pit_stops.csv
	qualifying.csv
	red_flags.csv
	safety_cars.csv
	seasons.csv
	sprint_results.csv
	virtual_safety_car_estimates.json

2.3 Tables

2.3.1 circuits.csv

This file contains information about each circuit where a Formula 1 race has been held.

Column	Description
circuitId	Unique identifier for each circuit
circuitRef	A short reference name for the circuit
name	The official name of the circuit
location	The city where the circuit is located
country	The country where the circuit is located
lat	Latitude of the circuit
lng	Longitude of the circuit
alt	Altitude of the circuit in meters
url	URL for the circuit's Wikipedia page

2.3.2 constructor_results.csv

This file contains the results for each constructor in each race.

Column	Description
constructorResultsId	Unique identifier for each constructor result
raceId	Foreign key to <code>races.csv</code>
constructorId	Foreign key to <code>constructors.csv</code>
points	Points scored by the constructor in the race
status	Status of the constructor in the race

2.3.3 constructor_standings.csv

This file contains the constructor standings after each race.

Column	Description
constructorStandingsId	Unique identifier for each constructor standing
raceId	Foreign key to <code>races.csv</code>
constructorId	Foreign key to <code>constructors.csv</code>
points	Total points for the constructor after the race
position	Constructor's position in the standings
positionText	Text representation of the constructor's position
wins	Number of wins for the constructor in the season

2.3.4 constructors.csv

This file contains information about each constructor.

Column	Description
constructorId	Unique identifier for each constructor
constructorRef name	A short reference name for the constructor
nationality	The full name of the constructor
url	The nationality of the constructor
	URL for the constructor's Wikipedia page

2.3.5 driver_standings.csv

This file contains the driver standings after each race.

Column	Description
driverStandingsId	Unique identifier for each driver standing
raceId	Foreign key to races.csv
driverId	Foreign key to drivers.csv
points	Total points for the driver after the race
position	Driver's position in the standings
positionText	Text representation of the driver's position
wins	Number of wins for the driver in the season

2.3.6 drivers.csv

This file contains information about each driver.

Column	Description
driverId	Unique identifier for each driver
driverRef number	A short reference name for the driver
code	The driver's car number
forename	A three-letter code for the driver
surname	The driver's first name
dob	The driver's last name
nationality	The driver's date of birth
url	The driver's nationality
	URL for the driver's Wikipedia page

2.3.7 races.csv

This file contains information about each race.

Column	Description
raceId	Unique identifier for each race
year	The year of the race
round	The round number of the race in the season
circuitId	Foreign key to <code>circuits.csv</code>
name	The name of the Grand Prix
date	The date of the race
time	The time of the race
url	URL for the race's Wikipedia page
fp1_date	Date of Free Practice 1
fp1_time	Time of Free Practice 1
fp2_date	Date of Free Practice 2
fp2_time	Time of Free Practice 2
fp3_date	Date of Free Practice 3
fp3_time	Time of Free Practice 3
quali_date	Date of Qualifying
quali_time	Time of Qualifying
sprint_date	Date of Sprint Race
sprint_time	Time of Sprint Race

2.3.8 results.csv

This file contains the results for each driver in each race.

Column	Description
resultId	Unique identifier for each result
raceId	Foreign key to <code>races.csv</code>
driverId	Foreign key to <code>drivers.csv</code>
constructorId	Foreign key to <code>constructors.csv</code>
number	The driver's car number
grid	The driver's starting grid position
position	The driver's finishing position
positionText	Text representation of the driver's finishing position
positionOrder	The driver's finishing position in order
points	Points scored by the driver in the race
laps	Number of laps completed by the driver
time	The driver's total race time
milliseconds	The driver's total race time in milliseconds
fastestLap	The lap number of the driver's fastest lap
rank	The rank of the driver's fastest lap
fastestLapTime	The driver's fastest lap time
fastestLapSpeed	The speed of the driver's fastest lap
statusId	Foreign key to <code>status.csv</code>

2.3.9 status.csv

This file contains information about the finishing status of a driver in a race.

Column	Description
statusId	Unique identifier for each status
status	A description of the finishing status

2.4 Entity Relationship Diagram Model

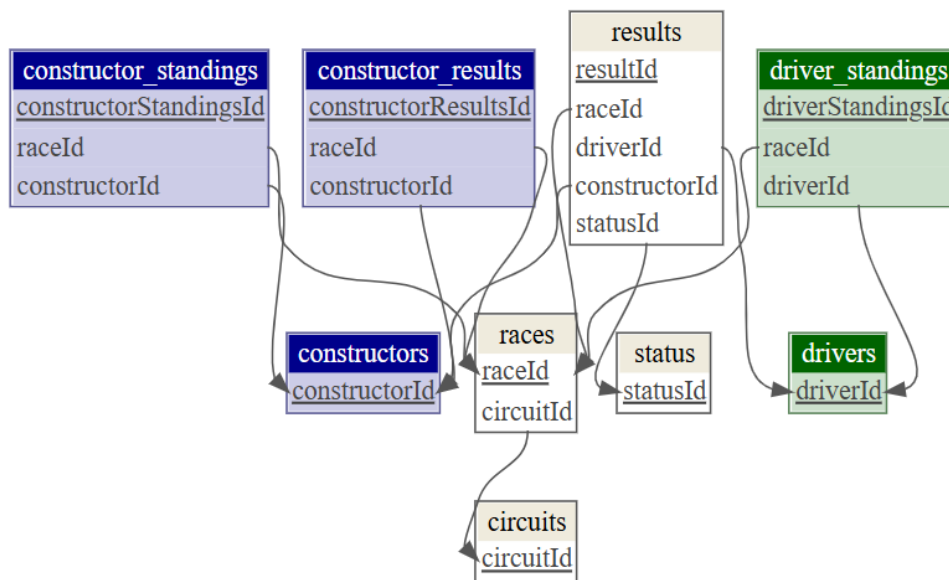


Figure 2: Entity Relationship Diagram Model

2.5 Data Issues

The following figure show that 14 rows in the **driver_standing** table have a position different to the positionText, with only one being “D” (which seems to be for Disqualified) :

driverStandingsId	raceId	driverId	points	position	positionText	wins
11031	223	30	78	26	D	5
73285	1145	859	0	\N	-	0
73286	1145	864	0	\N	-	0
73287	1145	4	0	\N	-	0
73288	1145	832	0	\N	-	0
73289	1145	862	0	\N	-	0
73290	1145	865	0	\N	-	0
73310	1146	4	0	\N	-	0
73785	1169	840	0	\N	-	0
73786	1169	4	0	\N	-	0
73787	1169	822	0	\N	-	0
73788	1169	865	0	\N	-	0
73789	1169	857	0	\N	-	0
73790	1169	807	0	\N	-	0

The raceId 223 and driverId 30 correspond to Michael Schumacher who was disqualified after the Grand Prix of Europe in 1997 :

raceId	year	name	driverId	forename	surname
223	1997	European Grand Prix	30	Michael	Schumacher

However this wasn’t the only disqualification from the history of Formula 1. The table below shows 5 rows from the results table containing 151 observations related to a disqualification (statusId of 2) :

resultId	raceId	driverId	constructorId	grid	position	positionText	positionOrder	points	laps	statusId
22	18	22	11	10	\N	D	22	0	58	2
499	41	13	6	5	\N	D	21	0	51	2
500	41	21	4	9	\N	D	22	0	51	2
1005	64	27	13	21	\N	D	21	0	66	2
1006	64	33	13	18	\N	D	22	0	66	2
1028	65	9	2	9	\N	D	22	0	69	2

More interestingly, the raceId 223 and driverId 30 from the results dataset don’t indicate a disqualification but a collision :

statusId	status
4	Collision

Now if we look at fatal accidents (statusId = 104), we can only find 3 results :

resultId	grid	positionOrder	laps	statusId	year	name	forename	surname	status
11303	23	25	0	104	1982	Canadian Grand Prix	Riccardo	Paletti	Fatal accident
14953	23	23	9	104	1974	United States Grand Prix	Helmuth	Koinigg	Fatal accident
16133	0	23	0	104	1970	Italian Grand Prix	Jochen	Rindt	Fatal accident

We know that more drivers died during a Formula 1 Grand Prix.

For example Ayrton Senna's status in the 1994 San Marin GP was marked as accident (statusId = 3) :

resultId	grid	positionOrder	laps	statusId	year	name	forename	surname	status
5216	1	22	5	3	1994	San Marino Grand Prix	Ayrton	Senna	Accident

Which can indicate that the fatal accidents only report drivers who died on-site during a race or qualifications.

The following table shows a list of statuses, only the first 12 are displayed :

statusId	status
1	Finished
2	Disqualified
3	Accident
4	Collision
5	Engine
6	Gearbox
7	Transmission
8	Clutch
9	Hydraulics
10	Electrical
11	+1 Lap
12	+2 Laps

We will only want to have the following statuses : "Finished", "Abandoned", "Lapsed" and "Disqualified". The "Lapsed" meaning the driver finished the race but with 1 or more laps behind the winner. The "Abandoned" includes all sort of mechanical problems and collisions, accidents... The "Disqualified" status is the same as "Abandoned" but the driver then didn't score any point.

The only exception found is Stirling Moss during the 1959 French GP who scored one point (best lap). He was disqualified during the race and kept his only point for the best lap :

resultId	points	year	name	forename	surname	status
18372	1	1959	French Grand Prix	Stirling	Moss	Disqualified

3. Data Transformation

We will now preprocess the Formula 1 dataset, focusing on optimizing the data frames for analysis by removing unnecessary columns, joining relevant data, renaming columns for clarity, and calculating driver age.

3.1 Removing Unnecessary Columns

To streamline the dataset, unnecessary columns from various data frames are removed. The goal is to retain only relevant information that will contribute to the overall analysis.

Circuits: The columns lat, lng, and url are omitted from the circuits data frame as they do not provide essential information for the analysis of race results.

Constructor Results & Standings: In the constructor_results and constructor_standings data frames, the status and positionText columns are excluded, respectively, as they are not required for further analysis.

Drivers' Columns: Columns related to drivers, such as number and url in the drivers data frame, are removed from driver_standings, simplifying the dataset.

Races Table: The races data frame is streamlined to include only the columns raceId, year, round, circuitId, name, and date, ensuring relevant race information is retained while discarding less useful details.

Results' Columns: A significant reduction in the results data frame is performed by dropping columns such as number, time, milliseconds, as well as fastest lap information (fastestLap, fastestLapTime, fastestLapSpeed), which may not be needed for the analysis of race outcomes.

3.2 Modifying statuses

As noted in the previous section, the status table includes various statuses that may not be relevant to this study. We will update these to reflect more accurate values.

3.3 Joining Tables

To create a comprehensive Formula 1 dataset, the results data frame is merged with several other data frames using left joins based on common identifiers.

The results data frame is join together with races, drivers, constructors, and status, linking them through raceId, driverId, constructorId, and statusId, respectively. This process allows for the integration of race and participant information into a single dataset.

3.4 Finalizing the Dataset

After merging the data frames, additional columns that are no longer necessary (raceId, driverId, constructorId, circuitId) are excluded to reduce clutter in the dataset.

3.5 Renaming Columns

To enhance clarity and facilitate a better understanding of the dataset, several columns are renamed. The following renaming conventions are employed:

name.x becomes **circuit name** **name.y** becomes **constructorName** **nationality.x** becomes **driverNationality** **nationality.y** becomes **constructorNationality**

These changes improve readability and usability of the dataset for analysis.

3.6 Calculating Driver Age

Driver age is computed by converting the date and dob columns into Date format. The age is calculated using the difference in weeks between the two dates. The result is then divided by 52.25 to convert weeks into years, and the floor() function is applied to round down to the nearest integer. This new column, driverAge, is added to the dataset.

3.7 Concatening Driver Names

The driver's full name is created by concatenating the forename and surname. The original two columns are then removed from the dataset.

3.8 Cleaning Up Date Columns

Finally, the date and dob columns are removed from the dataset, as they are no longer necessary after calculating driver age.

Overall, these transformations enhance the dataset's efficiency and layout, preparing it for insightful analysis in subsequent sections of the report.

3.9 Formula 1 Dataset

The dataset displayed below is a snippet of the final transformed dataset, showcasing only 8 out of the 16 columns for enhanced clarity:

resultId	positionOrder	points	year	circuit	driverName	driverAge	constructorName
1	1	10	2008	Australian Grand Prix	Lewis Hamilton	23	McLaren
2	2	8	2008	Australian Grand Prix	Nick Heidfeld	30	BMW Sauber
3	3	6	2008	Australian Grand Prix	Nico Rosberg	22	Williams
4	4	5	2008	Australian Grand Prix	Fernando Alonso	26	Renault
5	5	4	2008	Australian Grand Prix	Heikki Kovalainen	26	McLaren
6	6	3	2008	Australian Grand Prix	Kazuki Nakajima	23	Williams

The formula1 dataset contains the following columns :

Column	Description
resultId	Unique identifier for each result
grid	The driver's starting grid position
positionOrder	The driver's finishing position in order
points	Points scored by the driver in the race
laps	Number of laps completed by the driver
rank	The rank of the driver's fastest lap
statusId	Foreign key to <code>status.csv</code>
year	The year of the race
round	The round number of the race in the season
circuit	The official name of the circuit
driverNationality	The driver's nationality
constructorName	The full name of the constructor
constructorNationality	The nationality of the constructor
status	A description of the finishing status
driverAge	The driver's age at the time of the race
driverName	The driver's full name

4. Exploratory Data Analysis

4.1 Top Drivers

4.1.1 Names

If we define the leading drivers as those with the highest number of titles, and among those who are world champions with an equal number of titles, we can use the number of wins to differentiate them. This results in the following table displaying the 17 drivers who won at least 2 titles (as of 2025) :

driverName	titles	wins	races	seasons	points	titlespc	winspc
 Lewis Hamilton	7	105	383	20	4,990.50	0.350	0.274
 Michael Schumacher	7	91	308	19	1,566.00	0.368	0.295
 Juan Fangio	5	24	58	8	279.14	0.625	0.414
 Max Verstappen	4	71	236	12	3,313.50	0.333	0.301
 Sebastian Vettel	4	53	300	16	3,098.00	0.250	0.177
 Alain Prost	4	51	202	13	798.50	0.308	0.252
 Ayrton Senna	3	41	162	11	614.00	0.273	0.253
 Jackie Stewart	3	27	100	9	360.00	0.333	0.270
 Niki Lauda	3	25	174	13	420.50	0.231	0.144
 Nelson Piquet	3	23	207	14	485.50	0.214	0.111
 Jack Brabham	3	14	129	16	261.00	0.188	0.109
 Fernando Alonso	2	32	431	23	2,380.00	0.087	0.074
 Jim Clark	2	25	73	9	274.00	0.222	0.342
 Mika Häkkinen	2	20	165	11	420.00	0.182	0.121
 Emerson Fittipaldi	2	14	149	11	281.00	0.182	0.094
 Graham Hill	2	14	179	18	289.00	0.111	0.078
 Alberto Ascari	2	13	36	6	140.00	0.333	0.361

For each driver, we observe the number of titles, total races won, total races competed, the number of seasons or years they participated, the total points earned (only those that count toward the championship), the ratio of titles to seasons, and the ratio of wins to races.

Although the number of titles and races won clearly shows that Michael Schumacher and Lewis Hamilton outshine other drivers, the same is not necessarily the case for points. When we rank drivers by points, both Verstappen and Vettel surpass Schumacher. Even Alonso who won 2 titles, compared with the 7 by Schumacher, has more points than the latter.

To add another layer of complexity, if we look at the ratio of titles to seasons or the ratio of wins to races, Juan Manuel Fangio emerges as the top performer. Even Ascari, who appears at the bottom of the list, has the second best ratio win to races.

Now if we look at the first 9 pilots, only 2 of them (Fangio and Stewart) were racing at the earliest days of Formula 1. On the opposite, if we look at the last 8 drivers, 4 of them were driving at the earliest days : Brabham, Clark, Hill and Ascari. We can also notice that most of these early drivers had a career under 10 years, when modern days drivers have a career that last more than 10 years if not 20. Only notable exception is Graham Hill who had a long career of 18 years from 1958 to 1975.

We will now see the careers and titles of these 17 drivers across the years.

4.1.2 Timeline

The plot below illustrates the duration and timeline of careers without indicating any breaks; for instance, Alain Prost did not compete in 1992, and Schumacher was absent from the tracks from 2007 to 2009.

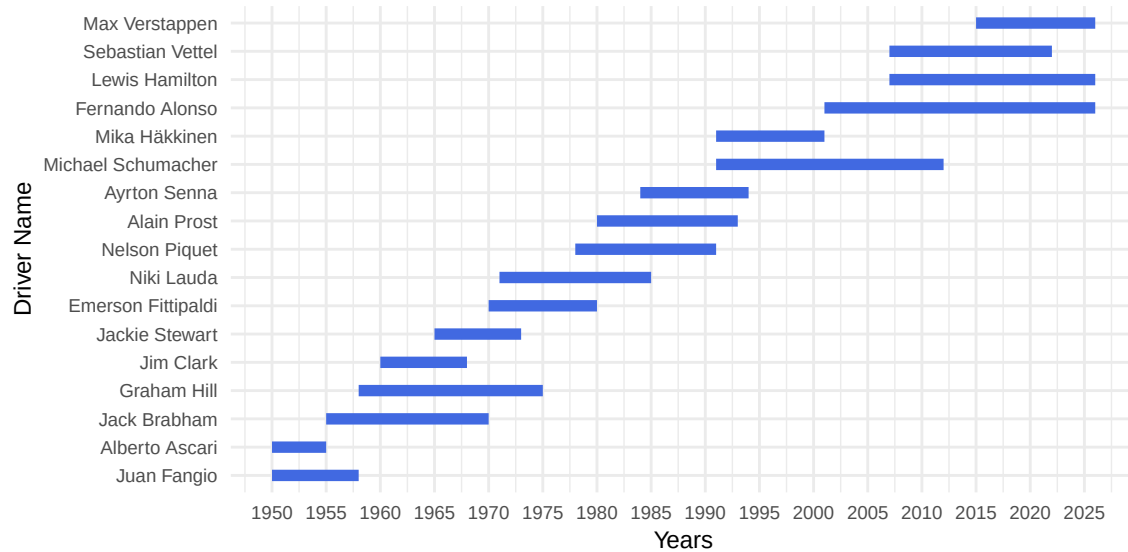


Figure 3: Careers

Some drivers enjoyed longer careers than others. For instance, Alonso, Schumacher, and Hamilton had the longest career spans, while Ascari, Clark, and Stewart had the shortest. The problem with this graph is that it does not effectively depict the dominance of any particular driver. For instance, Alonso secured his two titles in 2005 and 2006, and has continued racing since without winning any additional titles.

Now, if we replace the career duration with titles, where each segment represents the years between a driver's first and last title, we obtain the following graph :

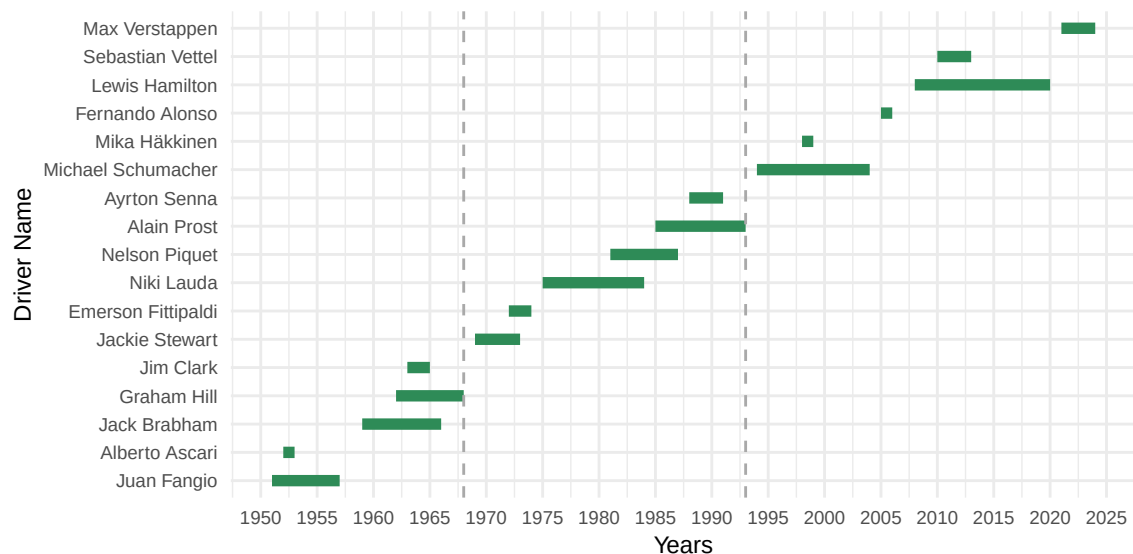


Figure 4: Titles

We divided this timeline into three phases: the first phase, the early stage, extends from the beginning until the late 1960s. The second phase begins with Jackie Stewart's first title in 1969 and continues until Michael Schumacher's first title in 1994. The final phase, or modern era, starts in 1994 and continues to the present.

In the plot above, each phase is marked with a vertical dashed grey line.

Now, when we display the career lengths based on the year of the first title, we can observe a gradual increase over the years :

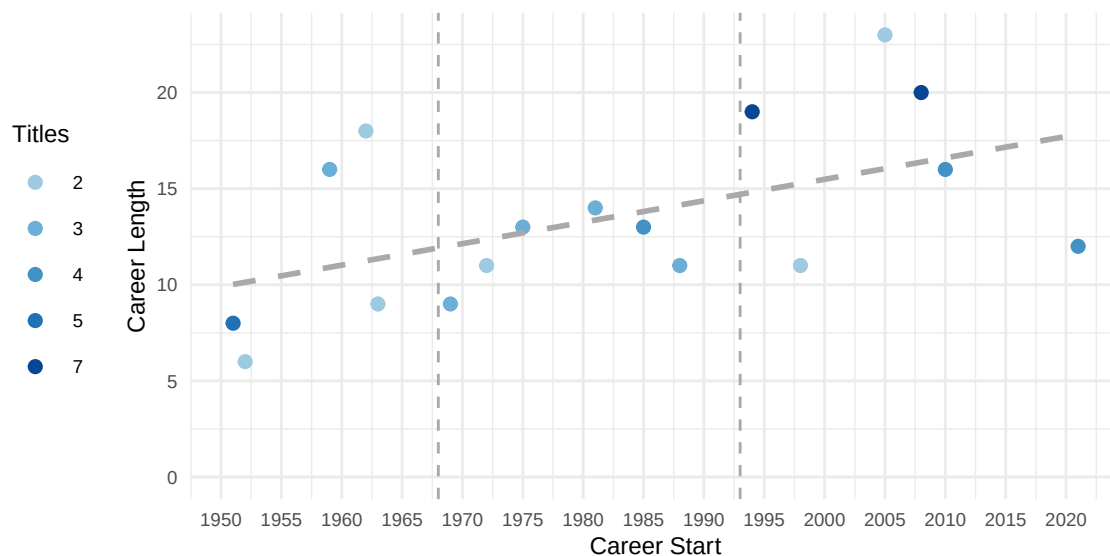


Figure 5: Career Start vs Career Length

We can notice that while Fangio was able to score 5 titles within 8 years in the 1950s, Schumacher and Hamilton scored their 7 titles in 19 and 20 years (with the latter still competing).

Let's now focus on the number of wins and races per driver :

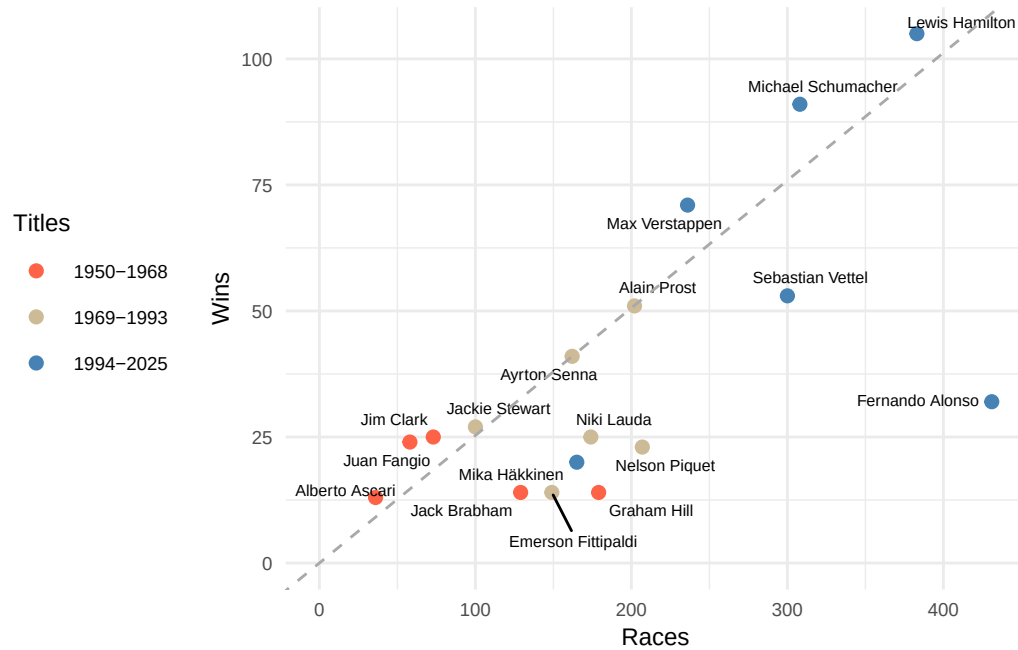


Figure 6: Wins vs Races

There is a noticeable upward trend in the plot, suggesting that drivers who participate in more races tend to achieve a higher number of wins. This correlation implies that experience and track time significantly contribute to success in the sport. Drivers from the 1994 to 2023 era, such as Hamilton and Vettel, are positioned in the top right quadrant, indicating both a high number of races and wins. The representation of this era in a distinct color underscores the advantages of modern technology and greater competitiveness, reflecting a notable shift in performance standards.

The varying colors used to group drivers by era emphasize the differences in victory rates and opportunities for success across time. Historical drivers may have achieved fewer wins but did so in a context with fewer races and different competitive landscapes. For instance, earlier eras, represented in warmer colors, often had fewer wins despite high title counts, pointing to the unique challenges faced by those drivers. Additionally, Fernando Alonso stands out for having a considerable number of races with only two titles, highlighting that a lengthy career does not guarantee championship success. This suggests that various external factors, such as team dynamics and competitive context, play significant roles in determining outcomes.

Overall, the plot illustrates that while participation in a greater number of races can lead to more wins, it does not exclusively dictate a driver's success. Factors like team performance, car technology, and the competitiveness of each era are crucial. The shifts in driver success across different decades indicate an evolution in Formula 1, influenced by changes in race formats and team strategies, as well as an expanding global audience. The color-coded groupings serve to highlight how drivers' experiences and achievements are shaped by the specific challenges and opportunities of their respective eras. This analysis provides a thorough understanding of the factors influencing driver success in Formula 1, as visualized through the plot, while also acknowledging the historical context that contributes to these differences.

And now the number of titles and seasons per driver :

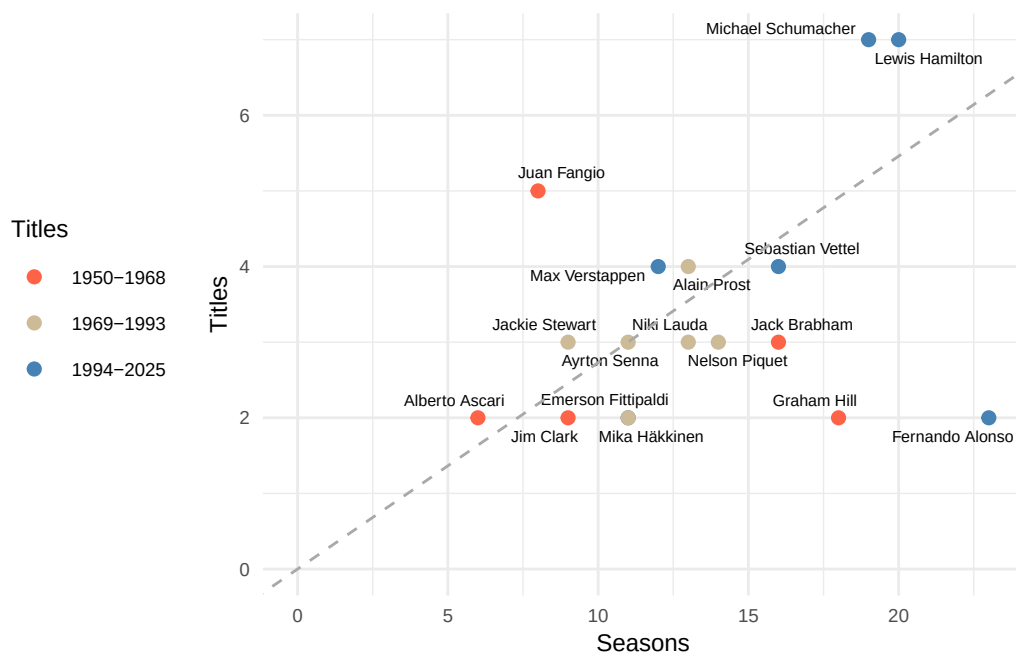


Figure 7: Titles vs Seasons

While the plot displays a trend where drivers with more seasons tend to win multiple titles, this is true for those who have achieved two or more titles. Notably, modern drivers such as Hamilton and Vettel, positioned in the top right quadrant, exemplify this trend, showcasing the advantage of extended careers combined with high performance. Their achievements in more recent years highlight the advancements in technology and team strategies that have influenced competitive outcomes.

The color coding of the data points emphasizes differences in performance across historical eras. For instance, drivers from the 1950 to 1988 timeframe, such as Fangio and Clark, achieved multiple titles within fewer seasons, reflecting the unique challenges of their competitive environments, which often had fewer races and different dynamics compared to today. This juxtaposition illustrates how the number of titles can vary significantly based on the era in which the drivers competed.

Additionally, the representation of drivers like Alonso showcases the relationship between seasons and titles, as he has spent a considerable number of seasons in the sport but has only two titles. This highlights that having a long career does not always correlate with championship victories, suggesting the impact of factors like team performance, adaptability, and the competitive landscape at any given time.

Overall, the plot underscores that while participation in more seasons can lead to multiple titles, it does not guarantee success. Various factors, including team dynamics, technological advancements, and the competitive environment in different eras, play crucial roles in determining a driver's achievements. The shifts in performance across different decades indicate an evolution within Formula 1, influenced by changing race formats and an increasingly global audience. This analysis provides a comprehensive understanding of how the number of seasons impacts title success while also recognizing the historical context that shapes these outcomes.

9. References

- [1] Hergé. 1956. “The Calculus Affair” <https://www.tintin.com/en/albums/the-calculus-affair>
- [1] Jean-Luc. 2019. “Tintin – The Calculus Affair – The liberties of English translation” <https://tintinomania.com/tintin-affaire-tournesol-anglais-brabham>
- [2] Kaggle “Formula 1 World Championship (1950 - 2024)” <https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>
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- [8] Cover image from SVG SILH <https://svgsilh.com/c3c3c3/image/309862.html>
- [9] Icons from Font Awesome <https://fontawesome.com/>
- [10] The text of this report was reviewed by DuckAI, an AI language model that also assisted in reviewing and correcting some of my code : <https://duck.ai/>